

EXERCISE SHEET 9

GROUP AND RING THEORY

MAIN QUESTIONS - TO BE DONE BEFORE THE SEMINAR

9.1. Let R be a ring. An R -module M is *almost finitely cogenerated* if it is not finitely cogenerated but all its proper quotients are finitely cogenerated.

Suppose M is an almost finitely cogenerated R -module. For any $f \in \text{End}_R(M)$, if $f = 0$ or $\ker f = 0$, and if $f(M) \neq M$, show that $\bigcap_{n \geq 1} f^n(M) = 0$.

9.2. Let R be a Noetherian integral domain, and let $0 \neq c \in R$. Show that $Rc \cap A \neq 0$ for all non-zero left ideals A in R . [Hint: If $Rc \cap A = 0$, then $\sum_{n=0}^{\infty} Ac^n$ is a direct sum.]

Does the result still hold upon removing the condition that R is an integral domain?

9.3. Let R be a left Artinian integral domain with more than one element. Show that R is a division ring.

EXTRA QUESTIONS

9.4. Let M be an R -module. Prove that the following are equivalent:

- (i) M is Artinian;
- (ii) every quotient module of M is finitely cogenerated;
- (iii) every non-empty set S of submodules of M has a minimal element (that is, a submodule M_0 in S such that for any submodule N_0 in S with $N_0 \subseteq M_0$ we have $N_0 = M_0$).

9.5. Let M be an R -module. If, for any (not necessarily countable) chain of non-zero submodules of M , the intersection of all the terms of the chain is non-zero, show that M is finitely cogenerated.

9.6. Show that a left Artinian ring cannot possess an infinite direct sum $\bigoplus A_i$ of left ideals A_i of R .

9.7. Show that the matrix ring over an Artinian ring is also Artinian.

9.8. Let R be a ring. Recall that an *idempotent* is an element $a \in R$ such that $a^2 = a$. For $R = \mathbb{Z}/n\mathbb{Z}$, show that a is an idempotent if and only if $a(a - 1) = 0$. Then, in $R = \mathbb{Z}/6\mathbb{Z}$, find all idempotents.